

Solar Energy Power Plants in California – A Binary Classification Approach

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1 General introduction

This dataset contains information about solar power plants and their various geographic and technical attributes. The dataset supports energy planning, renewable energy development, and spatial analysis, providing insights into the distribution and characteristics of solar projects in California. It is particularly valuable for utility companies, policy analysts, and researchers focused on renewable energy infrastructure. The dataset consists of 5397 rows and 20 columns (excluding the OBJECTID index column). We have missing values on four columns, especially related to unique identifiers (HIFLD IDs) for certain substations (e.g., HIFLD ID (GTET 100 Max Voltage), HIFLD ID (GTET 200 Max Voltage) and HIFLD ID (CAISO)). The following columns are given:

1. **County:** The name of the Californian county where the data point is located, represented as a string.
2. **Acres:** The area in acres ($1 \text{ ac} = 4046.8564224 \text{ m}^2$) associated with each data point, represented as a float.
3. **Install Type:** The type of solar installation (modes: 'Rooftop', 'Parking', 'Ground'), represented as a string.
4. **Urban or Rural:** A classification of whether the location is in an urban or rural area (modes: 'Urban', 'Rural'), represented as a string.
5. **Combined Class:** A combination of the install type and the column "Urban or Rural" (modes: 'Rooftop - Urban', 'Parking - Urban', 'Ground - Urban', 'Ground - Rural', 'Rooftop - Rural', 'Parking - Rural'), represented as a string.
6. **Distance to Substation (Miles) GTET 100 Max Voltage:** The distance from the data point to a substation with a GTET 100 Max Voltage, in miles ($1 \text{ mi} = 1,609.344 \text{ m}$), represented as a float.
7. **Percentile (GTET 100 Max Voltage):** The percentile ranking related to the GTET 100 Max Voltage distance, represented as a string.
8. **Substation Name GTET 100 Max Voltage:** The name of the substation with a GTET 100 Max Voltage, represented as a string.
9. **HIFLD ID (GTET 100 Max Voltage):** A unique identifier for the GTET 100 Max Voltage substation, represented as a float, though some values are missing.
10. **Distance to Substation (Miles) GTET 200 Max Voltage:** The distance from the data point to a substation with a GTET 200 Max Voltage, in miles ($1 \text{ mi} = 1,609.344 \text{ m}$), represented as a float.
11. **Percentile (GTET 200 Max Voltage):** The percentile ranking related to the GTET 200 Max Voltage distance, represented as a string.
12. **Substation Name GTET 200 Max Voltage:** The name of the substation with a GTET 200 Max Voltage, represented as a string.
13. **HIFLD ID (GTET 200 Max Voltage):** A unique identifier for the GTET 200 Max Voltage substation, represented as a float, though some values are missing.

14. **Distance to Substation (Miles) CAISO:** The distance from the data point to a CAISO substation, in miles (1 mi = 1,609.344 m), represented as a float.
15. **Percentile (CAISO):** The percentile ranking related to the CAISO substation distance, represented as a string.
16. **Substation CASIO Name:** The name of the CAISO substation, represented as a string, though a few values are missing.
17. **HIFLD ID (CAISO):** A unique identifier for the CAISO substation, represented as a float, though some values are missing.
18. **Solar Technoeconomic Intersection:** A string indicating the technoeconomic intersection related to solar energy (modes: 'Within', 'Outside'), referring to areas with high or low solar potential or feasibility.
19. **Shape__Area:** The area of the geographical shape in squaremetres, represented as a float.
20. **Shape__Length:** The length of the geographical shape, represented as a float.

Key Features:

- Location information, including county and urban/rural classification.
- Proximity to high-voltage substations (≥ 100 kV and ≥ 200 kV) and CAISO substations.
- Solar installation type (e.g., rooftop) and area in acres.
- Percentile rankings for distances to substations.
- Spatial details, including shape area and length for GIS applications.

Target:

The **Solar Technoeconomic Intersection** column suggests that this data may be used for analysis related to solar energy generation. This column could be used as target variable because it classifies the area's suitability for solar technology.

This dataset could be valuable for analyzing energy infrastructure in relation to geographic regions, urban/rural areas, and distances to various types of substations, potentially useful for planning, optimization, and analysis in energy systems or grid management.

Key sources:

1. Data: <https://www.kaggle.com/datasets/vijayveersingh/california-sunny-spaces>
2. Data visualized with interactive map: <https://gis.data.cnra.ca.gov/datasets/CAEnergy::solar-footprints-in-california/about>

2 Overview and Preliminary Data Exploration

The dataset under consideration provides crucial insights into the characteristics of solar power plants across California, focusing on key geographic and technical attributes. The data spans multiple features, including the type of solar installations, proximity to various substations, and the feasibility of solar energy projects in different areas based on their technoeconomic potential. By analyzing this data, the project aims to shed light on the distribution of solar energy infrastructure and assist in the optimization and development of energy systems.

Initial data processing involved cleaning and exploring the dataset using various Python libraries, with a focus on handling missing values, managing categorical and numerical variables, and preparing the dataset for deeper analysis. The process of understanding the structure and quality of the data is crucial for any further exploration and modeling.

2.1 Data Cleaning and Preprocessing

The first step in the data analysis was examining the basic structure and attributes of the dataset. The dataset contains 5397 rows and 20 columns, with several key attributes that provide insights into the solar energy infrastructure in California. Some of the notable features include Install Type (such as 'Rooftop', 'Parking', 'Ground'), Urban or Rural classification, and distance-related attributes to substations.

A detailed review of the dataset revealed some important issues:

1. **Missing Values:** Four columns related to substation identifiers (e.g. HIFLD ID) had missing values. The share of missing values was below 10% for those columns. Because of the lack of informative value for the modeling these columns were dropped.
2. **Data Redundancy:** Some columns, such as the Shape__Area column, were redundant as they provided the same information as other columns (e.g., Acres), but in different units. These redundant columns were dropped to streamline the dataset.
3. **Long Column Names:** To simplify data processing and visualization, long column names, particularly those related to substations, were shortened using an abbreviation dictionary. For example, columns like "Distance to Substation (Miles) GTET 100 Max Voltage" were renamed to "Distance to GTET 100" for ease of access and clarity.
4. **Additional columns:** Because of the geographical nature of the dataset it was plausible to add more geographical information like latitude and longitude as well as the population density as columns to the dataset. This was achieved by using the geopandas library and open source internet resources.

Finally, the target variable “Solar Technoeconomic Intersection” was converted to binary values (1 for "Within" and 0 for "Outside") to facilitate subsequent statistical and machine learning analysis. This step ensured that the data was in an appropriate format for modeling.

2.2 Exploratory Data Analysis (EDA)

Summary Statistics

After cleaning the dataset, an initial exploration of the numerical features was performed. Summary statistics were generated to better understand the distribution and spread of key attributes like Acres, Distance to Substations, and pop_density. These statistics provided a useful overview of the data's central tendencies and variability, highlighting any outliers or skewed distributions.

Missing Values Analysis

The missing values across the dataset were systematically analyzed to identify the proportions of missing data in each column. Some columns, especially those related to substation identifiers, exhibited significant missing data. This is important because these missing values could affect downstream analyses, such as any spatial or proximity-based assessments related to the distribution of substations and solar installations.

Duplicate Data Detection

The dataset was also examined for duplicated entries. It was confirmed that there were no significant duplicates present, which ensures the integrity of the dataset for further analysis.

Categorical Variables Analysis

A categorical feature analysis was performed for columns such as Install Type, Urban or Rural, and Combined Class. The unique values of these features were reviewed to gain insights into the distribution of solar installations across urban and rural areas, as well as to determine the relative frequency of different installation types (rooftop, parking, or ground).

Correlation Analysis

A correlation matrix was computed to examine the relationships between numerical features. This step was instrumental in identifying potential relationships between key attributes, such as the correlation between the distance to substations and other geographic factors.

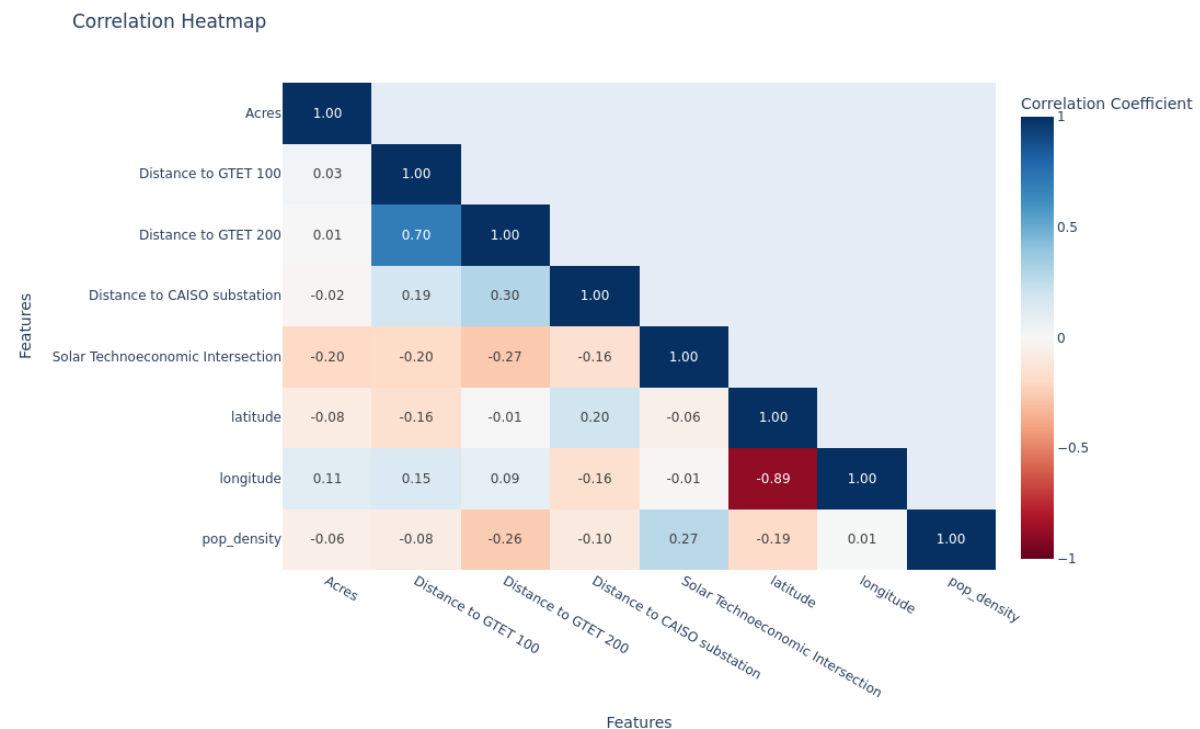
2.3 Data Visualization

Key visualizations were created to further explore and communicate the data's insights.

Correlation Heatmap

A heatmap visualization was generated to represent these correlations visually, allowing for a clear identification of positive or negative relationships. The heatmap indicated interesting

trends, such as the inverse relationship between the distance to substations and the potential solar technoeconomic viability of different locations. This provides an initial indication that areas closer to substations may have better feasibility for solar installations, due to easier grid access. The strongest correlation with the target variable are given with “pop_density” (0.27) and “Distance to GTET 200” (-0.27).

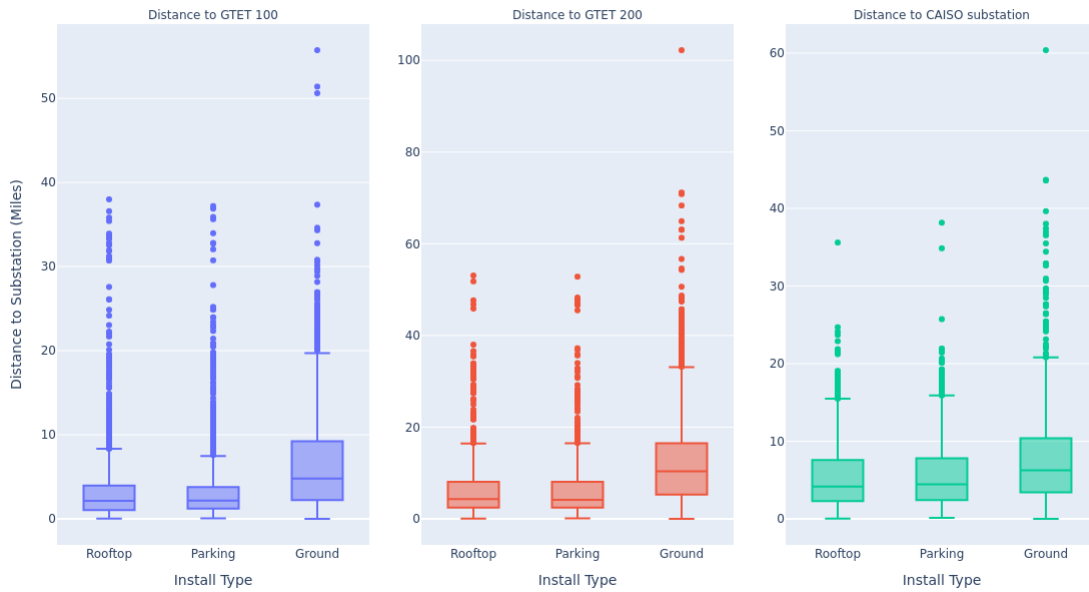


The distance features (to substation GTET 100, GTET 200 and CAISO) show some moderate to weak correlations with each other, with the strongest being between the distance to GTET 100 and to GTET 200 (0.70). The geographic features in the dataset like longitude and latitude are very strongly negatively correlated with each other which is because of the elongated, rectangular shape of the state of California which stretches from north west to south east. The distance measures like GTET and CAISO have a weak correlation whereas using domain knowledge (i.e. here the "business" perspective) it is understandable that the distances of the solar power installations to the different substation types indicate that installations closer to one GTET location tend to be closer to the other as well.

Boxplot Matrix

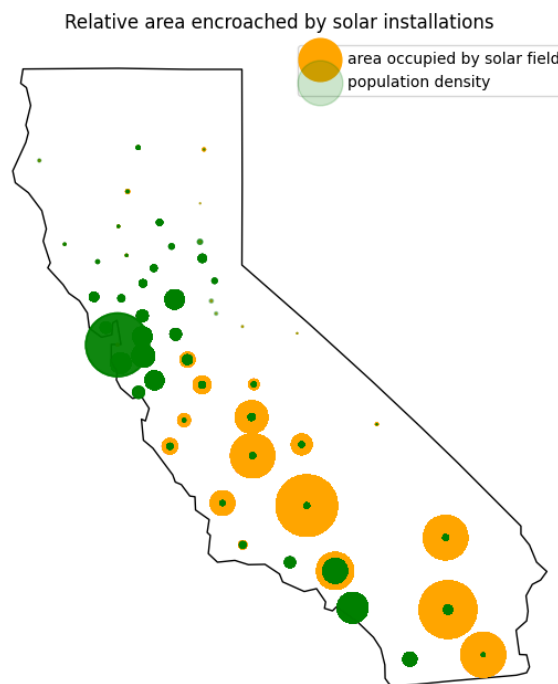
A boxplot matrix was created to compare the distances to various substations (GTET 100, GTET 200, and CAISO) by installation type. This boxplot provided an intuitive view of how different installation types are distributed in terms of their proximity to high-voltage substations. The results showed that Rooftop installations tend to be closer to substations compared to Ground installations, which typically have more variable distances (outliers).

Boxplot: Distance to Substations by Install Type and Substation Type



Geospatial Analysis

One of the key aspects of this dataset is the geographic distribution of solar installations across California counties. To support this, the geolocation of each county was retrieved using the geopy library. While the geocoding process encountered some rate-limiting issues due to the number of requests, it was planned that this step would allow for the addition of latitude and longitude information for further spatial analysis. These visualizations allowed for a better understanding of the geographic patterns of solar installations in California and their proximity to critical energy infrastructure.



2.4 Statistical Analysis and Tests

We explored several relationships between the target variable ("Solar Technoeconomic Intersection" – a binary variable) and other features in the dataset, using appropriate statistical tests. The goal was to assess if there were significant associations between the target and other variables, including continuous, categorical, and binary features.

1. Point-Biserial Correlation: Relationship between Pop Density and Solar Technoeconomic Intersection

Since "Pop Density" is a continuous variable and "Solar Technoeconomic Intersection" is binary (0 or 1), we used Point-Biserial Correlation to assess the strength and direction of the association between these variables. Before proceeding with the correlation, we verified whether "Pop Density" follows a normal distribution using the Shapiro-Wilk test for normality. The result showed that "Pop Density" is not normally distributed ($p\text{-value} < 0.05$), which led us to apply a non-parametric test.

The Point-Biserial correlation (essentially a special case of Pearson's correlation) is a suitable test for measuring the strength and direction of the association between a binary variable and a continuous variable. The Point-Biserial Correlation was computed, with a correlation coefficient and $p\text{-value}$ indicating the strength of the relationship. If the $p\text{-value}$ is below the significance threshold (0.05), it would imply a significant relationship. The result showed that there was no significant correlation between "Pop Density" and the target variable ($p\text{-value} > 0.05$).

2. Point-Biserial Correlation: Relationship between Distance to GTET 200 and Solar Technoeconomic Intersection

Next, we assessed the relationship between "Distance to GTET 200" (a continuous variable) and the binary target variable. Similarly to the previous analysis, we used the Shapiro-Wilk test to check for normality. The results indicated that "Distance to GTET 200" was not normally distributed. Consequently, we proceeded with the Point-Biserial Correlation, yielding a $p\text{-value}$ greater than 0.05. This result suggested that there is no significant relationship between the distance to the GTET 200 and the Solar Technoeconomic Intersection.

3. Chi-Square Test of Independence: Relationship between Install Type and Solar Technoeconomic Intersection

One can use also the Chi-Square test of independence to check if there's a relationship between the categorical variables and a binary target. While the correlation coefficient is often used to measure the strength of association, the Chi-Square test helps testing whether the proportions of a binary target differ by category (e.g., do "Urban" and "Rural" categories have different proportions of $\text{target}=1$ or $\text{target}=0$).

Thus, for categorical variables, we used the Chi-Square Test of Independence to test if there was a significant association between "Install Type" (a categorical variable) and "Solar Technoeconomic Intersection." The contingency table for the two variables was created, followed by the Chi-Square test. The test statistic and p-value were computed to determine if the variables were independent or associated. The p-value from this test (which was below 0.05) indicated a significant relationship between "Install Type" and the target variable.

4. Chi-Square Test of Independence: Relationship between Urban or Rural and Solar Technoeconomic Intersection

We applied the Chi-Square Test of Independence once again, this time to explore the relationship between "Urban or Rural" (a categorical variable) and "Solar Technoeconomic Intersection." A contingency table was constructed for these variables, and the Chi-Square test was performed. The p-value of the test suggested a significant relationship between "Urban or Rural" and the target variable (p-value < 0.05), meaning that the classification of "Urban or Rural" could influence the target variable.

Conclusion of the statistical analysis

Through these statistical tests, we were able to evaluate key relationships between the target variable and other variables in the dataset. While the Point-Biserial Correlation tests showed no significant relationships for the continuous variables, the Chi-Square tests revealed significant associations between the categorical variables ("Install Type" and "Urban or Rural") and the target. The visualizations further aided in understanding these relationships.

2.5 Interim conclusion

As interim conclusion, the preliminary steps of data cleaning, exploration, and visualization have laid a solid foundation for the project's next stages. The dataset's rich geographic and technical details offer a promising opportunity to advance the analysis of solar energy projects, with significant implications for energy infrastructure planning and renewable energy development in California.

3 Pre-Model preparation

3.1 Data Split and Encoding

To ensure robust model performance, the dataset was split into training and testing sets, with the target variable "Solar Technoeconomic Intersection" classifying areas as "Within" or "Outside." The training and testing split was 80% and 20% respectively, with the testing set preserving the original class distribution.

Categorical features in the dataset were encoded to convert them into a numerical format suitable for modeling. Percentile-related variables ("Percentile (GTET 100)," "Percentile (GTET 200)," and "Percentile (CAISO)") were ordinally encoded using predefined categories ("0 to 25th," "25th to 50th," "50th to 75th," "75th to 100th") to preserve their inherent order, creating new columns ("percentile_GTET100," "percentile_GTET200," "percentile_CAISO"). The original percentile columns were then dropped.

The "Install Type" variable was one-hot encoded using dummy variables, with the "Ground" category dropped to avoid multicollinearity and the dummy variable trap, ensuring no redundant information for linear models. This resulted in binary columns ("Parking" and "Rooftop") for the remaining categories.

The "Urban or Rural" and "Solar Technoeconomic Intersection" variables were binary encoded, mapping "Urban" to 0 and "Rural" to 1, and "Within" to 1 and "Outside" to 0, respectively, to create integer representations. The "County" variable was dropped due to its high cardinality and lack of informative value for the prediction. Finally, the dataset was reordered, placing the target variable as the last column for clarity.

3.2 Sampling Strategies

Two sampling strategies were tested to address class imbalances (70.23% of the training data has a Solar Technoeconomic Intersection = 1, while only 29.77% have a Solar Technoeconomic Intersection = 0) and maintain data representativeness: oversampling and undersampling.

The first strategy, random oversampling, targeted the minority class to balance the dataset. Synthetic examples of the minority class were generated, increasing its representation in the training set. The testing set remained unchanged to reflect the original class distribution, ensuring realistic evaluation. Oversampling was valuable for enhancing model performance on the minority class, critical for identifying suitable solar deployment areas.

The second strategy, undersampling with cluster centroids, reduces the size of the majority class without randomly removing data, ensuring that the representative centroids of the majority class are kept, which can lead to more efficient training and better performance in some cases. While this approach also addressed class imbalance, it risked losing valuable information.

After evaluation, oversampling was selected as the final approach. In our case it provided a more robust solution for handling class imbalances without sacrificing data, ensuring the model generalizes well across both classes.

3.3 Scaling Strategies

Feature scaling was applied to standardize numerical features, such as "Acres," distances to substations, and "Population Density," which vary widely in scale. Two scaling strategies were tested: min-max scaling and robust scaling.

The first strategy, min-max scaling, transformed features to a fixed range, typically [0, 1], by using the `MinMaxScaler` function of Scikit-learn. Scaling parameters were computed from the training data and applied to both training and testing sets to prevent data leakage. While min-max scaling is effective for features with known bounds, it is highly sensitive to outliers. Extreme values in features like "Acres" and distances to substations could compress the majority of the data into a narrow range, reducing the effectiveness of the scaling.

The second strategy, robust scaling, used the median and interquartile range (IQR) to transform features, making it less sensitive to outliers. This method subtracts the median and divides by the IQR (difference between the 75th and 25th percentiles), with parameters computed solely from the training data to ensure no leakage. Robust scaling preserved feature relationships despite extreme values, making it particularly suitable for this dataset, where outliers were present in several numerical features.

After evaluation, robust scaling was chosen as the final approach. It handled outliers more effectively than min-max scaling, ensuring fair feature contributions to models sensitive to magnitudes, such as logistic regression or neural networks. Categorical features, already encoded, were not scaled.

3.4 Data Export

Once the sampling and scaling processes were complete, the preprocessed data was exported to ensure accessibility and reproducibility for subsequent analysis and modeling phases. The training and testing sets (`X_train`, `X_test`, `y_train`, `y_test`) were saved as separate files in a structured format, facilitating their use by other team members and enabling seamless integration into the broader project workflow. The data was exported in CSV format, ensuring compatibility with various analytical tools and platforms.

To maintain clarity and organization, the files were named descriptively, such as "`X_train.csv`," "`X_test.csv`," "`y_train.csv`," and "`y_test.csv`," and stored in a dedicated directory. After this step the exported data is ready for use in the modeling phase, where it will support the development of predictive models to assess solar technoeconomic suitability.

4 Modeling

4.1 Classification of the problem

4.1.1 Type of Machine Learning Problem

This project is a supervised learning problem that falls under the category of classification. Specifically, it is a binary classification problem where the goal is to predict the Solar Technoeconomic Intersection variable, which refers to areas with high or low solar potential or feasibility (modes: 'Within', 'Outside').

4.1.2 Task Relevance

Our task is closely related to the energy sector, supporting energy planning, renewable energy development, and spatial analysis. It is particularly valuable for utility companies, policy analysts, and researchers focused on renewable energy infrastructure.

4.1.3 Performance Metric

The primary performance metric chosen for this problem is the f1-score. This metric is selected because it balances precision and recall, making it ideal for imbalanced datasets.

Additional metrics used include precision and confusion matrix, as they provide deeper insights into model performance and help fine-tune the decision threshold.

ROC Curve

Another performance evaluation method we used is the Receiver Operating Characteristic (ROC) curve which is a graphical representation that shows the predictive performance of a binary classifier across all classification thresholds. It plots the True Positive Rate (TPR) against the False Positive Rate (FPR). A key metric derived from the ROC curve is the Area Under the Curve (AUC), which quantifies the classifier's performance. An AUC of 1.0 indicates perfect performance, while an AUC of 0.5 suggests that the model performs no better than random guessing. Although AUC alone doesn't capture all aspects of model performance (such as precision or recall), it is a good single-number summary of overall model performance.

4.2 Model choice and optimization

4.2.1 Algorithms Tried

We trained four classifier models: RandomForest, SVM (Support vector machine), XGBoost, and KNN (k nearest neighbors). To optimize hyperparameters in various machine learning algorithms, we employed GridSearchCV. This method trains the model with all possible combinations of the specified parameter grid and utilizes cross-validation to evaluate each combination. The optimal combination is selected based on a performance metric, which in our case is the f1-score.

4.2.2 Model Selection

We used Grid-search to optimize parameters for the best f1-score with a cross-validation parameter of 5.

To expand the scope of the optimization search, the GridsearchCV procedure was repeated on three training datasets: a) an oversampled (OS) training dataset scaled using a min-max scaler, b) an oversampled training dataset scaled using a robust scaler, and c) an undersampled (US) training dataset scaled using a robust scaler.

4.2.3 Parameter Optimization

To determine the most effective model with optimized parameters, the following table presents the f1-score performances of various models.

f1-scores after Gridsearch		OS, min-max scaling	OS, robust scaling	US, robust scaling
RandomForest	Train	1.0000	0.9983	0.9794
	Test	0.9506	0.9390	0.9446
SVM	Train	0.8976	0.9250	0.8884
	Test	0.9261	0.9153	0.9282
XGBoost	Train	0.9997	0.9997	0.9674
	Test	0.9451	0.9299	0.9376
KNN	Train	1.0000	1.0000	1.0000
	Test	0.9335	0.9359	0.9341

The RandomForest algorithm, trained on oversampled and min-max scaled data, yields the highest f1-scores on test datasets. However, the table shows many train datasets with an f1-score of 1, indicating overfitting. To determine which model is more generalized, we calculate the ratio of the test f1-score to the train f1-score. These values are presented in the following table.

f1-scores after Gridsearch	OS, min-max scaling	OS, robust scaling	US, robust scaling
RandomForest	0.9506	0.9406	0.9645
SVM	1.0318	0.9895	1.0448
XGBoost	0.9454	0.9302	0.9692
KNN	0.9335	0.9359	0.9341

The above ratios simply show how well a model performs with the test dataset compared to the train dataset. Higher the ratio more generalized the model is. With this criteria SVM model has worked the best, yet we discard the model trained on oversampled-min-max-scaled and undersampled-robust-scaled train data due their low train f1-scores.

4.2.4 Conclusion

With above analysis, we choose SVM with parameters C = 10, gamma = auto, and kernel = rbf as the best classifier model.

4.3 Interpretation of results

4.3.1 Error Analysis

The classification report of the best model selected is:

```
Classification Report:
              precision    recall  f1-score   support

     0           0.83       0.90       0.86         322
     1           0.96       0.92       0.94         758

 accuracy              0.91         1080
 macro avg           0.89       0.91       0.90         1080
 weighted avg        0.92       0.91       0.92         1080
```

The confusion matrix for training data set:

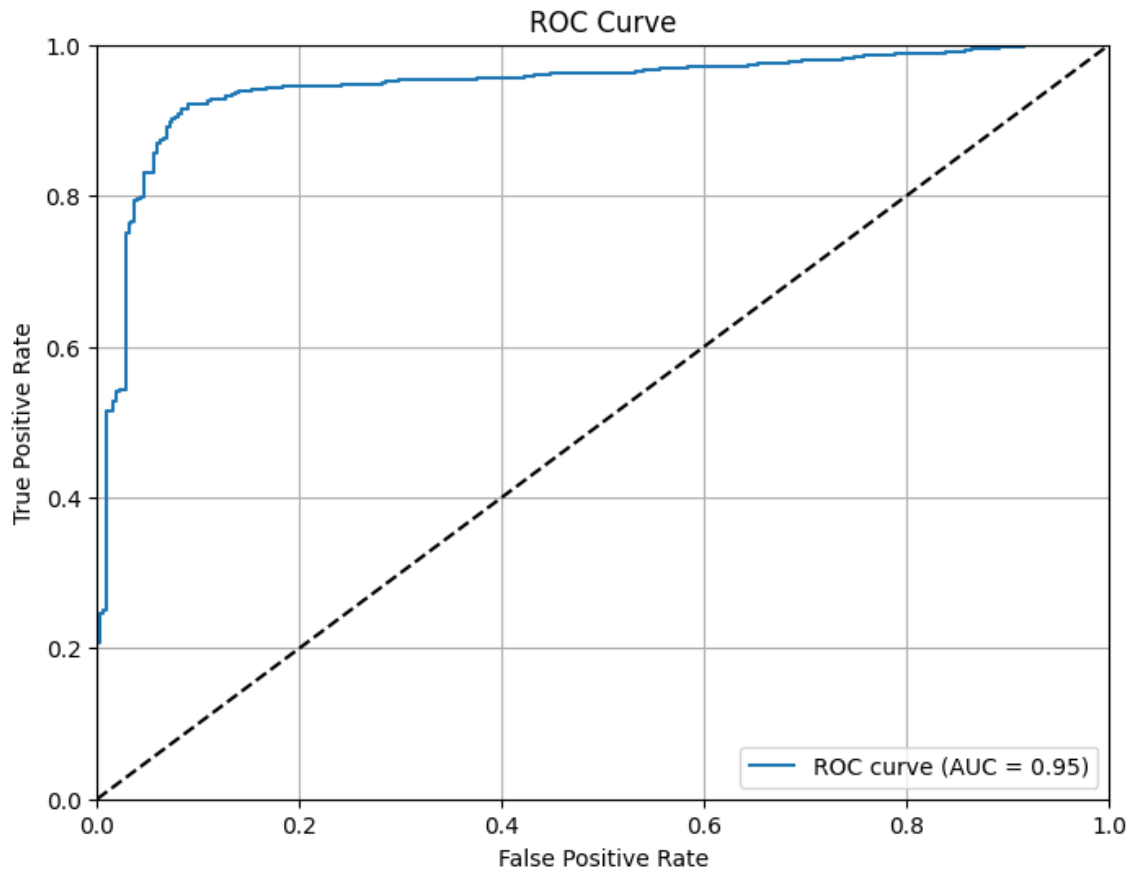
	1	0
1	2845	187
0	239	2793

The confusion matrix for test data set:

	1	0
1	290	32
0	60	698

The following steps were taken to plot the ROC curve of the SVC (Support Vector Classification) model:

1. Predicted Probabilities: The `predict_proba()` method is used to get the predicted probabilities for the test set, specifically the probabilities for the positive class (class 1).
2. ROC Curve Calculation: The false positive rate (FPR) and true positive rate (TPR) are calculated using the `roc_curve()` function.
3. AUC Calculation: The ROC AUC score is computed using `roc_auc_score()` to measure the model's performance.



The resulting plot visualizes the trade-offs between sensitivity (TPR) and specificity ($1 - \text{FPR}$), providing insight into the model's classification performance. A diagonal dashed line representing random classification ($\text{AUC} = 0.5$) is also included for reference. With our resulting value of $\text{AUC} = 0.95$ we reached a strong predictive performance with our model which suggests that the model is very good at distinguishing between the two classes (positive/within the solar technoeconomic intersection and negative/outside of the solar technoeconomic intersection). The ROC curve also rises sharply towards the top-left corner which indicates that the model achieves high TPRs with low FPRs across a wide range of threshold settings.

4.3.2 Interpretability Techniques

After evaluating the models with GridSearchCV and selecting the best performing one, model interpretability is conducted using a key visual technique: SHAP (SHapley Additive exPlanations).

SHAP Summary Plot

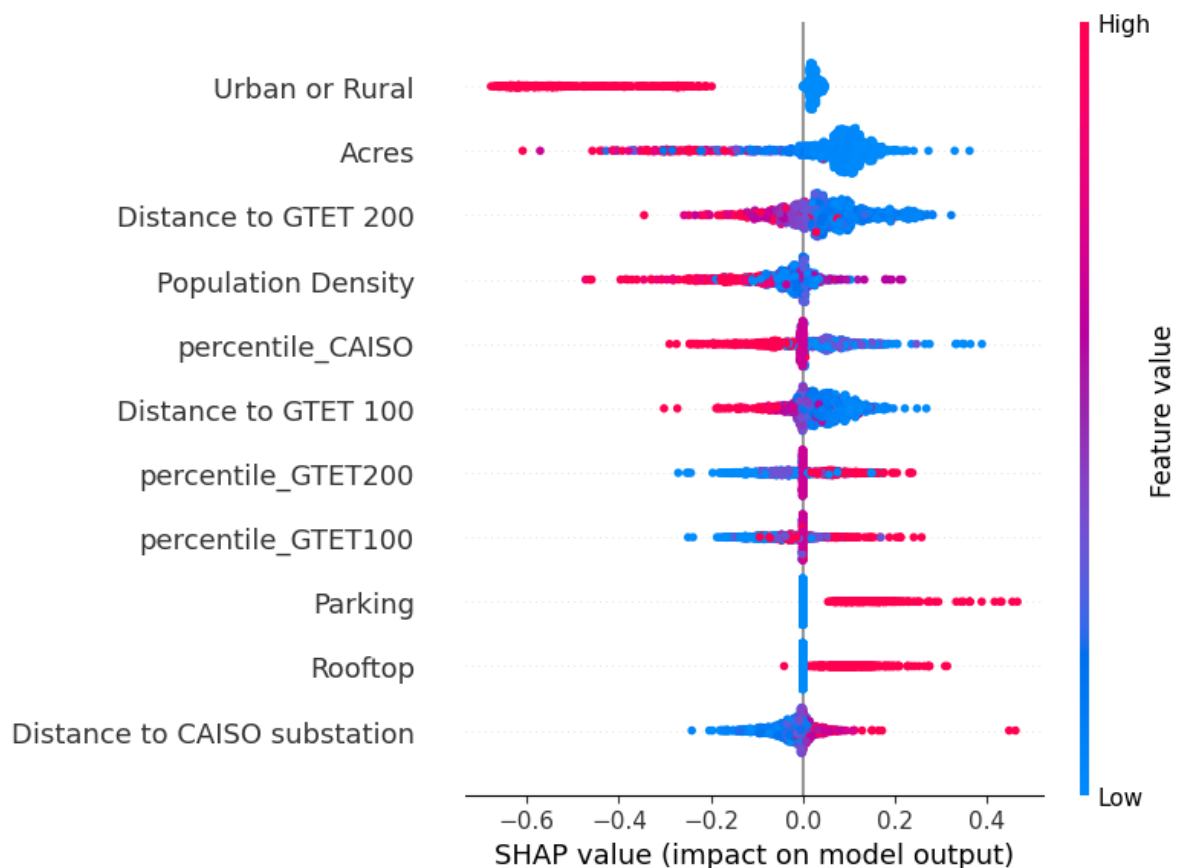
SHAP (SHapley Additive exPlanations) is a model-agnostic technique used to interpret the predictions made by machine learning models. It assigns each feature an importance score, indicating how much each feature contributes to the model's output for a given prediction. This is particularly useful for understanding complex models.

We performed the following steps to generate a SHAP summary plot:

1. **Background Data:** To reduce computational cost, K-means clustering is used to select a smaller set of representative background samples (reduced to 10 samples in this case). This step helps speed up SHAP computation, as it reduces the size of the dataset that the SHAP algorithm needs to process.
2. **SHAP Explainer:** A KernelExplainer is initialized with the model's `predict_proba()` method and the background data. The explainer calculates SHAP values for each sample in the test set.
3. **SHAP Values:** The SHAP values are computed for the test set, which indicate the contribution of each feature to the prediction.
4. **Summary Plot:** The SHAP summary plot provides a global interpretation of the model. It shows the distribution of SHAP values for each feature, helping to understand the relative importance of each feature and the direction of its impact on the model's predictions. Features with higher SHAP values are considered more influential.

The resulting plot (see next section) helps in understanding the model's interpretability, showing which features are most important for predicting the target variable, and whether they have a positive or negative influence on the model's output.

4.3.3 Key Insights



Urban or Rural: This feature appears to be the most important predictor in the model. However, the concentration of red dots (high values - representing rural areas) on the negative SHAP value side suggests that being in a rural area negatively impacts the model's output (lower probability of solar energy feasibility).

Acres: The second most important feature is "Acres." Here, the cluster of blue dots on the positive SHAP value side suggests that lower acreages tend to positively impact the model's prediction. This could indicate that smaller properties are more likely to be suited for solar energy.

Distance to GTET 200 ("GTET" refers to "Grid-Connected Transformer Equipment Terminal"): Lower distances on this feature tend to have a positive impact on solar feasibility according to the SHAP values in the plot.

Population Density: Higher population density seems to have in particular a negative impact on the model's output. This means that, according to the SVC model, in areas with high population density, the probability of solar energy feasibility decreases.

percentile_CAISO: This feature has a distribution of both high and low values contributing negatively and positively to the SHAP value, depending on if the value is low or high. This feature has more impact to the model's output than the corresponding percentile_GTET100 or percentile_GTET200.

Distance to GTET 100: This feature shows some effect by showing an increase for the model output when this distance decreases.

percentile_GTET100 and percentile_GTET200: These percentile features seem to have both weak positive and negative impacts depending on their values.

Rooftop and Parking: These features have a positive SHAP value. "Parking" suggests that installations on parking lots are more likely to be feasible for solar energy. Higher values lead to a positive impact on the model's output. Similarly, "Rooftop" has a positive influence.

Distance to CAISO substation: According to the SHAP values, this feature has the lowest effect on the model output.

4.4 Evaluation of the model reliability

4.4.1 Bias and Fairness Considerations

For a bias free and a fair model performance, we resampled the train dataset. To that end, We employed random oversampling and cluster-centroid undersampling techniques. To ensure equal contributions from different variable, we scaled the training and test datasets using robust and min-max scalers. For model training, we use three sets of data: a) an oversampled (OS) training dataset scaled using a min-max scaler, b) an oversampled training dataset scaled

using a robust scaler, and c) an undersampled (US) training dataset scaled using a robust scaler.

4.4.2 Robustness Testing

The model performances on train and test datasets were comparable, which ensured the robustness of the model. Hence, a detailed robustness test by adding random noise to the different variables was not carried out.

4.4.3 Deployment Readiness

The model is performing well, but as mentioned below we will conduct more tests before making it deployment ready. In our Preprocessing notebook, we experimented with a pipeline that currently only includes the scaling process. This pipeline could be further enhanced to incorporate the entire workflow, including data splitting, encoding, resampling, scaling, and prediction etc.

4.5 Optimization proposal

4.5.1 Potential Improvements

Additionally, we tried the implementation of a machine learning model using a stacking classifier, which combines multiple base models to improve predictive performance. In our case, the process includes hyperparameter optimization using Optuna and employs nested cross-validation to evaluate model performance and optimize hyperparameters. This also integrates various machine learning algorithms with a meta-learner to ensure improved performance. And finally, SHAP (SHapley Additive exPlanations) is used for model interpretability, allowing insights into the feature importance and model decisions.

The goal was to construct a robust and high-performing classification model capable of generalizing well to unseen data but since the process had its limitations (time and score) we decided for now not to implement this method for our modeling. Nevertheless, it is useful to explain how this approach works in the following because this approach could be pursued in the future to optimize our model further.

4.5.2 Stacking Classifier Overview

A stacking classifier is an ensemble learning method where predictions from multiple base models are combined using a meta-learner to make the final prediction. The base models used in this implementation are:

- Random Forest Classifier (RF)
- Support Vector Classifier (SVC)
- K-Nearest Neighbors (KNN)
- XGBoost (XGB)

The predictions from these base models serve as input to the meta-model, which in this case is a Logistic Regression model. We decided for a simple meta-model because it is best practice and eases the interpretability. The stacking approach leverages the strengths of different classifiers and aims to improve overall performance by combining them.

The use of base models that differ in their underlying algorithms (e.g., decision trees, support vector machines, and gradient boosting) increases the diversity of the ensemble, which often leads to better generalization compared to individual models.

4.5.3 Hyperparameter Tuning with Optuna within the Stacking Trial

To enhance the performance of the base models, Optuna is used for hyperparameter tuning. Optuna is a framework that automates the search for the best hyperparameters using a trial-based optimization method.

The inner loop of the nested cross-validation process focuses on tuning the hyperparameters for each base model. The models have key hyperparameters that can be optimized:

- Random Forest (RF): The number of estimators (`n_estimators`) is tuned.
- Support Vector Classifier (SVC): The regularization parameter (`C`) and the kernel coefficient (`gamma`) are adjusted.
- K-Nearest Neighbors (KNN): The number of neighbors (`n_neighbors`) and leaf size (`leaf_size`) are optimized.
- XGBoost (XGB): The number of boosting rounds (`n_estimators`) and the learning rate (`learning_rate`) are tuned.

Optuna's objective function evaluates the model's performance by fitting a stacking classifier with the suggested hyperparameters and computing the validation accuracy. The optimization process is guided by MedianPruner, which prunes trials that show poor performance early, thus accelerating the search. But in our case, it had its application limitations because our search was not exhausting enough (e.g. low number of trials selected) because of the high time consumption. But with changing parameters this method would be very useful so that we decided to keep it.

4.5.4 Nested Cross-Validation for Stacking Model Evaluation

To evaluate the performance of the stacking classifier and its base models, nested cross-validation is used. This method helps in assessing the model's ability to generalize, while ensuring that hyperparameter optimization does not lead to overfitting. The nested cross-validation consists of two primary loops here:

- Outer Loop: The outer loop splits the dataset into training and validation sets multiple times (using `StratifiedKFold`). This ensures that the model is tested on different subsets of the data, providing a robust estimate of generalization performance.

- Inner Loop: Inside each outer fold, the inner loop is responsible for performing hyperparameter optimization via Optuna. The model's performance is assessed on a validation set, and the best hyperparameters are selected.

Both loops use StratifiedKFold, which ensures that the class distribution is maintained across each fold, e.g. important for classification tasks with imbalanced datasets. The inner loop tunes the hyperparameters on the training set of each outer fold, while the outer loop evaluates the final model's performance on the validation set.

4.5.5 Stacking Model Training and its Final Evaluation

Once the hyperparameters are optimized in the inner loop, the stacking classifier is trained using the best hyperparameters on the entire training dataset (from each fold of the outer loop). After training, the model is evaluated on the validation set, and its performance metrics (accuracy and ROC-AUC score) are recorded.

To assess the final model's performance, it is retrained with the best hyperparameters across all outer folds and then tested on the hold-out test set. This final evaluation provides key metrics such as accuracy, ROC-AUC score, confusion matrix, and classification report, offering a comprehensive overview of the model's performance on unseen data.

4.5.6 Interpretability with SHAP

Model interpretability is a crucial aspect, especially in complex ensemble methods like stacking. We used again SHAP to provide interpretability by calculating the contribution of each feature to individual predictions.

The SHAP analysis starts by using k-means clustering to summarize the background data, reducing the computational complexity. We used k=1 to reduce the time amount to several minutes instead of hours. The KernelExplainer from SHAP is used to compute SHAP values for the test set, which represent the impact of each feature on the model's predictions. A summary plot is generated to visualize the distribution of SHAP values, highlighting the features that contribute most to the model's decision-making.

4.5.7 Conclusion of the Stacking Trial

The described stacking model is a powerful but complex machine learning approach combining the capabilities of several base classifiers through a stacking ensemble. By optimizing the hyperparameters of each model using Optuna and evaluating the performance through nested cross-validation, the model should achieve a higher generalization ability. With the use of SHAP we tried to further add transparency, allowing for a better understanding of how features influence predictions. However, the complexity and time consumption disadvantage lead to our final rejection of the stacking approach but it clearly stays as a possible way for potential optimization.